Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher chooses one. Exercise 8 is revisited later.

Exercise 6.1 · A collection of derivatives

Consider the functions

$$f(x) = 3x^4 - 2x + 7, \quad g(x) = (x^2 + 2)(x - 1), \quad h(x) = \frac{2x + 1}{x + 3}.$$

1. State the domain of each of these functions.
2. Calculate their derivatives, naming the rule used each time. For g, first keep the two terms of the product rule; for h, write the numerator before simplifying it.
3. Check the result concerning g by expanding its expression before differentiating.

Exercise 6.2 · A complete table

The function f is defined on [0; 4] by $f(x) = x^3 - 6x^2 + 9x + 1$. We seek its minimum and maximum on this interval.

1. Calculate and factorise $f\prime(x)$, then study its sign on [0; 4].
2. Draw the variation table of f indicating the values at the bounds and at the points where the derivative is zero.
3. Give the minimum and maximum of f on the interval, as well as all the x-coordinates where they are reached.

Exercise 6.3 · A quotient

For every strictly positive real number x, set $f(x) = x + \frac{4}{x}$.

1. Study the variations of f on]0; $+\infty$[. Set out the factors that determine the sign of the derivative.
2. Deduce the smallest value of $x + \frac{4}{x}$, then write an inequality valid for every $x > 0$. State the equality case precisely.

Exercise 6.4 · Compare two curves

Functions $f(x) = x^3$ and $g(x) = x$ are defined on $\mathbb{R}$. Let $\mathcal{C}_f$ and $\mathcal{C}_g$ their curves in a same coordinate system.

1. Determine the intersection points of these two curves.

2. Study their relative position on $\mathbb{R}$ choosing a suitable method. Interpret your result by the intervals where $\mathcal{C}_f$ is above or below $\mathcal{C}_g$.

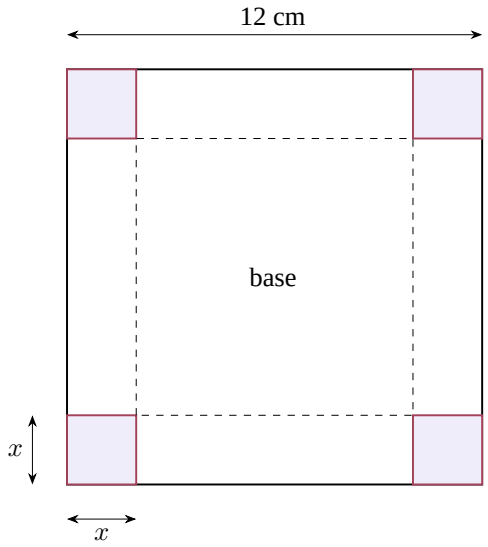
Exercise 6.5 · A package

An open box (no lid) is made from a square sheet of side 12 cm. We cut out in each of the four corners a square of side x cm, then the remaining edges are folded up. The thickness of the sheet is neglected.

1. Draw the net and express the height and the two dimensions of the base of the box. State the values of x which give a box of non-zero volume.

2. Express the volume $V(x)$ in cubic centimetres. Find the dimensions that make this volume maximum.

3. Check the values at both endpoints of the domain and give the exact maximum volume.



The four coloured areas are removed; the dotted lines show the folds. The drawing represents a illustrative value of x and does not give the optimum.

Exercise 6.6 · Stationary without extremum

A student claims: "If the derivative is zero at a point, the function must have an extremum local at this point." We study the function $f(x) = x^3$ to test this claim.

1. Calculate $f\prime(0)$.

2. Compare to $f(0)$ the values of $f(x)$ for real numbers x negative then positive, as close to 0 as we want.

3. Conclude about the existence of a local extremum at 0 and about the proposed statement.

Exercise 6.7 · Compare Strategies

We want to prove that, for every real number x, $x^2 - 6x + 12 \geq 3$.

1. Give a proof that makes a square appear.
2. Give a second proof by studying the variation of a function.
3. State the equality case. Compare the two methods: which one requires fewer calculations here?

Exercise 6.8 · Delayed review

Reciprocal function $f(x) = \frac{1}{x}$ has domain $D = \mathbb{R} \setminus \{0\}$. A student calculates $f'(x) = -\frac{1}{x^2} < 0$ and concludes that f is decreasing on the whole of D.

1. Identify the missing hypothesis in the application of the theorem linking derivative sign and changes.
2. State the two intervals on which the conclusion is correct.
3. Using the numbers -1 and 1, construct a counterexample to its conclusion on the whole D.

Independent work. Do only the task assigned at the end of the session, in the scheme of work Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not use up the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, Exercise 12 p. 156, Questions 1 and 2. Calculate the difference, study its sign and translate this into the relative position of the curves.

CQFD, exercise 16 p. 156, questions f and g. Give, for each function, an interval where it is both negative and increasing; justify both properties separately.